

F750 — Audit of K1026 (my own), against Cal’s decisive prior analysis (2026-07-15). Verdict: the sum rule is a genuine standard relation (NOT a tautology), but K1026’s “ $\cos 2\theta_{23} = -1/g \Rightarrow \theta_{23} = 4/7$ Derived” is NOT an independent forward prediction of θ_{23} — it rests on the δ -law ($\cos^2 \delta = N_c^2 n_C / g^2$) which is the SAME sum rule re-expressed, “true by construction” (Cal §2026-07-15). So K1026 does NOT override the shipped “maximal, 4/7 unforced” — it REFINES it: it identifies the deviation MECHANISM (the θ_{13} - δ back-reaction), but the deviation MAGNITUDE ($1/g$) is not independently forward-derived. No contradiction with the papers; θ_{23} stays MAXIMAL-Derived (parity theorem, leading order) with the sum rule as the δ -correlation, not a 4/7-forcing. The sharp audit question answered: the sum rule ($\cos 2\theta_{23} = \sin \theta_{13} \cdot \cos \delta$, Grimus–Lavoura, hep-ph/0601118) is one equation in three observables of ONE M_ν , so it yields ONE genuine forward output, not two — you cannot have BOTH $\theta_{23} = 4/7$ -Derived AND δ -Derived independently from one μ - τ -breaking parameter ε . The one genuine target-innocent output is the δ -PMNS prediction (given θ_{23}, θ_{13}); $\theta_{23} = 4/7$ is the same relation read backward, which is why it’s “true by construction,” not forward. The flip-condition (what WOULD make θ_{23} 4/7-Derived): an INDEPENDENT forward derivation of the δ -law $\cos^2 \delta = (N_c^2 \cdot n_C) / g^2$ — not via the sum rule. Until then, $\theta_{23} = \text{maximal}$.

Lyra, Wed 2026-07-30. Auditing my own K1026 with Keeper’s sharp question. Cal already worked this in July: the sum rule is standard, and the one forward output is δ — $\theta_{23}=4/7$ is the relation re-expressed. Owned: my “4/7-Derived” was over-optimistic. θ_{23} stays maximal; the papers are right.

The sharp question, answered (prediction vs tautology)

K1026 proposed: the Grimus–Lavoura sum rule $\cos 2\theta_{23} = \sin \theta_{13} \cdot \cos \delta$, with the derived $\theta_{13} = 1/45$ and a derived δ , gives $\cos 2\theta_{23} = -1/g = -1/7$ (the $\sqrt{5}$ ’s cancel: $\sin \theta_{13} = 1/\sqrt{45} = 1/(3\sqrt{5})$, $\cos \delta = -3\sqrt{5}/7$, product = $-1/7$) $\Rightarrow \sin^2 \theta_{23} = 4/7$. - The sum rule itself is STANDARD and target-innocent (Grimus–Lavoura, holds in ANY μ - τ -breaking model). Using it is not a fit. $\Rightarrow$ BUT (Cal §2026-07-15, the decisive point): the sum rule is ONE equation in three observables of ONE neutrino matrix, parametrized by ONE complex μ - τ -breaking ε (2 real DOF). So it delivers one genuine forward output, not two. “ $|\cos 2\theta_{23}| = 1/g$ being 4/7” and “ $\cos^2 \delta = 45/49$ ” and “ $|\varepsilon| = 1/\sqrt{45}$ ” are the same statement rearranged — TRUE BY CONSTRUCTION, not independent corroboration. - Therefore: $\theta_{23} = 4/7$ is genuine forward *only if* δ is independently derived (and θ_{13}); and $\delta = 49 = 45 + 4$ law ($\cos^2 \delta = N_c^2 n_C / g^2$) is itself the sum rule re-expressed with $\theta_{23} = 4/7$ as input. Circular as a θ_{23} -derivation. The one non-circular forward output is $\cos \delta / \delta$ -PMNS (given θ_{23}, θ_{13}). ■

Why this does NOT contradict the parity theorem (different sectors, leading + correction)

- Parity theorem (Cal, verified): the *condensate* forces the charged-lepton μ - τ block to maximal (μ mode odd, τ mode even, degree-1 condensate $\Rightarrow$ diagonal entries vanish $\Rightarrow \sin^2 \theta_{23} = 1/2$). This is the leading order, forced.

- K1026 sum rule: the *neutrino* $M_\nu \theta_{13}$ - δ back-reaction is a correction on top of the maximal leading order. Different sector — so it does not refute the parity theorem.
- Compatible reading: θ_{23} = maximal (leading, Derived) + a deviation (the back-reaction). The papers’ “maximal, 4/7 unforced” is exactly this: maximal is forced, the deviation magnitude is not. K1026 supplies the deviation MECHANISM (the sum rule) but not a forward MAGNITUDE — so it refines, doesn’t override.

Ruling

- θ_{23} = maximal (doubly-Derived: Shilov $\mathbb{Z}_2$ + parity theorem). Unchanged. The shipped papers are correct.
- The 4/7 deviation is Identified (mechanism = the Grimus–Lavoura back-reaction, magnitude NOT independently forward-derived).
- The sum rule’s genuine forward output is δ_{PMNS} — given θ_{23} (=observed/maximal) and θ_{13} (=1/45), it predicts $\cos\delta \approx 1/2$ ($\delta \approx 90^\circ$ (or the branch), DUNE-testable. That is the target-innocent prediction to report, NOT $\theta_{23} = 4/7$).
- K1026 is downgraded: not “ $\theta_{23} = 4/7$ Derived,” but “the μ - τ -breaking correlates θ_{23} , θ_{13} , δ ; the forward output is δ .” Owned — my sum-rule reading over-claimed θ_{23} , the same shape as F747/F749 (a relation read to give the pretty value rather than the forward one).

The flip-condition (the open computation, honest)

$\theta_{23} = 4/7$ becomes genuinely Derived iff the δ -law is independently forward-derived: $\cos^2\delta = (N_c^2 \cdot n_c)/g^2 = 45/49$ from the geometry, WITHOUT using θ_{23} . If someone derives δ (both magnitude and quadrant) forward from the imperfect- $\mathbb{Z}_2$ residual — Cal’s pre-registered bar (2026-07-15: “arg(ϵ) derived forward including quadrant”) — then $\theta_{13} + \delta$ + the sum rule over-determine $\theta_{23} = 4/7$, and it’s a genuine third prediction. That is a real open computation, not closed here.

Tier / handoffs

- @Grace — for the Falsifiable-Predictions paper: θ_{23} = maximal ($\sin^2\theta_{23} = 1/2$), doubly-Derived. Do NOT ship “ $\theta_{23} = 4/7$ Derived.” The sum rule is the δ_{PMNS} correlation ($\delta \approx 90^\circ$, DUNE-testable) — report THAT as the target-innocent output, with the θ_{23} deviation as an Identified correction (mechanism found, magnitude pending the δ -law). This is the material resolution your paper needed.
- @Cal — this audits K1026 against your own 2026-07-15 analysis: the sum rule is standard, one forward output ($\cos\delta$), “ $\cos 2\theta_{23} = 1/g$ ” is 4/7 re-expressed. Confirm the ruling (θ_{23} maximal; 4/7 Identified-not-forward; δ the genuine output), and own your δ_{PMNS} quadrant if it fed the confusion. The flip-condition (independent forward $\cos^2\delta = 45/49$) is the open piece.
- @Keeper — your sharp question resolved it: the sum rule is a genuine relation but relating three outputs of one M_ν gives ONE forward output, not θ_{23} -and- δ both. θ_{23} stays maximal; K1026 refines (mechanism) not overrides. Papers ship “maximal.” No block on the color-duality paper (θ_{23} only an example there).
- @Casey — resolved the one honest thread. My K1026 said the atmospheric angle is forced to 4/7 by a sum rule; auditing it (against Cal’s own July analysis) shows the sum rule is a real, standard relation but it only gives ONE genuine prediction from one imperfection — and that prediction is the CP phase δ , not the angle. The “4/7” is the same relation read backward with the observed angle plugged in, so it can’t also be a forward prediction of that angle — it’s true by construction. So θ_{23} stays *maximal*, exactly as the papers ship, and the sum rule’s real output is a $\delta \approx 90^\circ$ CP prediction for DUNE. No contradiction — my note refines (it found the deviation’s mechanism) but doesn’t force the number. It’s the same shape of slip I’ve owned twice this week (a relation read to give the pretty value instead of the forward one), and the discipline caught it before Grace’s paper finalized on the wrong answer. The one thing that WOULD force 4/7 is an independent derivation of δ from the geometry — a real open computation, honestly named, not closed. Nothing pushed; flagship still GO.

Notes only; no toy/theorem claimed. F750: AUDIT K1026 (mine) vs Cal §2026-07-15. Sum rule $\cos 2\theta_{23} = \sin\theta_{13} \cos\delta$ = STANDARD Grimus–Lavoura (target-innocent), but ONE eq in 3 observables of ONE M_ν (one ϵ) → ONE forward output. “ $\cos 2\theta_{23} = -1/g \rightarrow 4/7$ ” = the δ -law ($\cos^2\delta = N_c^2 n_c / g^2 = 45/49$) re-expressed = TRUE BY CON-

STRUCTION, NOT a forward θ_{23} derivation. Genuine forward output = δ_{PMNS} ($\approx 197^\circ$, DUNE). Does NOT contradict parity theorem (parity=charged-lepton LEADING maximal; sum rule= ν back-reaction CORRECTION). RULING: θ_{23} stays MAXIMAL (doubly-Derived); 4/7 deviation Identified (mechanism found, magnitude not forward); K1026 refines not overrides. Grace ships “maximal” + δ -correlation, NOT 4/7. Flip-condition: independent forward $\cos^2 \delta = 45/49$. Same slip-shape as F747/F749 (relation read backward for the pretty value). @Grace ship maximal. @Cal confirm + own δ quadrant. — Lyra